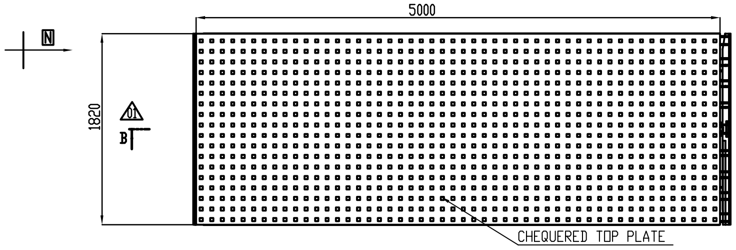
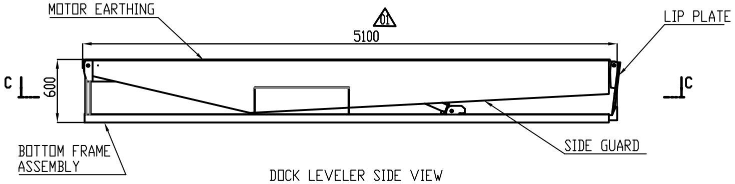
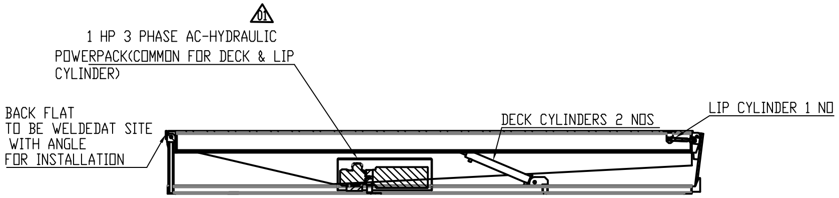


DOCK LEVELER TOP VIEW



DOCK LEVELER SIDE VIEW


△01
1 HP 3 PHASE AC-HYDRAULIC POWERPACK(COMMON FOR DECK & LIP CYLINDER)
BACK FLAT TO BE WELDEDAT SITE WITH ANGLE FOR INSTALLATION
DECK CYLINDERS 2 NOS
LIP CYLINDER 1 NO

SECTION B-B
SCALE 1 : 15

△01

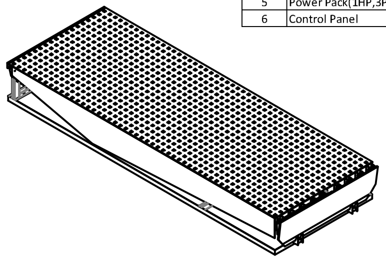
TECHNICAL SPECIFICATIONS OF PIT MOUNTED ELECRO-HYDRAULIC DOCK LEVELER -

1. SIZE OF DOCK LEVELER = 1820mm(W) X 5000mm(L)
2. OVERALL SIZE OF PRODUCT = 1820mm(W)X5100mm(L)X600mm(D)
3. CAPACITY = 6 TON POINT LOAD
4. LIP PLATE SIZE = 500mm(L)X1820mm(W)
5. DECK PLATE = CHEQUERED SHEET 1820mm(W)x5000mm(L)
6. DECK & LIP UP=DOWN OPERATION BY HYDRAULIC CYLINDER
7. DECK UP HEIGHT= (+)600 MM △01
8. DECK DOWN HEIGHT= (-)300MM △02
9. HEIGHT OF DOCK LEVELER FROM ROAD LEVEL= 900 mm,
10. OPERATION CONTROL BY PUSH BUTTONS ON CONTROL PANEL MOUNTED ON STAND NEAR DOCK LEVELER
11. ELECTRICAL SUPPLY OF EQUIPMENT= 415 VOLT 3 PHASE AC , 50HZ
12. PAINT = FIROZI BLUE
13. MATERIAL OF CONSTRUCTION = MILD STEEL AS PER IS STANDARD
14. MANUFACTURING STANDARD OF DOCK LEVELER =EN1398
15. ELECRICAL MOTOR COMPLIANCE ZONE 1-IIC-T3 HAZARDOUS AREA
16. PIT SIZE = 1870 mm(W)X5100mm(L)x600 mm(D)
17. DOCK LEVELER FIXING IN PIT BY ANCHOR BOLT .
18. REQUIRED PIT & ALL CIVIL WORK FOR MOUNTING OD DOCK LEVELER IS IN LTHE SCOPE .
REFER PIT DRAWING FOR CONSTRUCTION OF PIT B016-RUF-LT-504-MP-DW-NTPL(1)-13005
19. HYDRAULIC CYLINDERS & POWERPACK IS DESIGNED FOR LIFTING BARE DOCK ONLY.
20. MOTOR RATING=1HP 3 PHASE AC-HYDRAULIC
21. All ELECTRICAL EQUIPMENTS SHALL HAVE PAINT SHADE RAL 7031 △02
22. ALL FLAME PROOF CONTROL PANEL 375X375X230MM △05
23. WEIGHT= APPROX 2500KG EMPTTY & 8500KG OPERATING △02
24. WE WILL PROVIDE FRP CANOPY FOR LOCAL CONTROL PANEL. △02
25. FOR DOCK LEVELER PACKAGE LTHE SCOPE SHALL BE LIMITED TO PROVIDE SINGLE POINT SUPPLY (3 PHASE, 415 V,50 HZ) TO ISOLETOR PANEL LOCATED AT GRADE LEVEL. FURTHER ALL REQUIRED DISTRIBUTION SHALL BE IN VENDOE SCOPE. △03
26. REFER FOLLOWING TAG NUMBER FOR ELECTRICAL ITEMS.
MOTOR-504-LZ-930A-M1 & 504-LZ-930B-M1.
PANEL-LCP-504-LZ-930A & LCP-504-LZ-930B △03
27. LOCAL PANEL IS SUITABLE FOR ZONE1,IIC,T6 △05

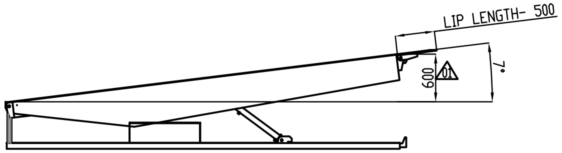
△02

BOM (For Single Dock Leveler)

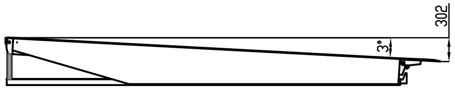
SL NO.	DESCRIPTION	QTY
1	Dock Leveler(1820m x 5000mm)	1No
2	Lip (1820 x500mm)	1No
3	Deck Cylinder	2Nos
4	Lip Cylinder	1No
5	Power Pack(1HP,3Phase)	1No
6	Control Panel	1No



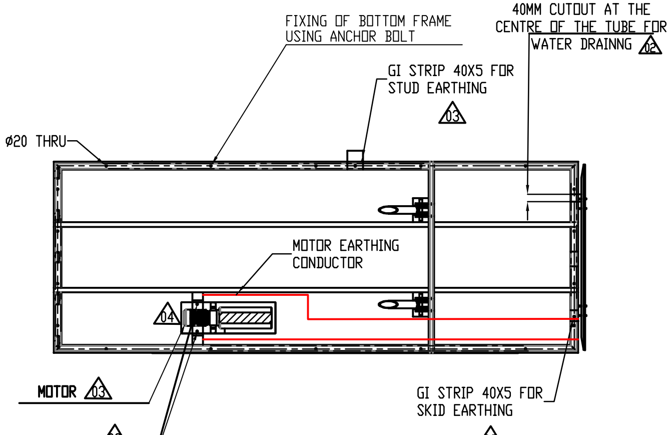
ISOMETRIC VIEW OF DOCK LEVELER



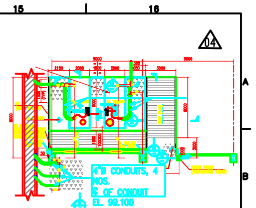
DECK UP HEIGHT - 600 MM



DECK DOWN HEIGHT - 300 MM


FIXING OF BOTTOM FRAME USING ANCHOR BOLT
40MM CUTOUT AT THE CENTRE OF THE TUBE FOR WATER DRAINNG △02
GI STRIP 40X5 FOR STUD EARTHING △03
Ø20 THRU
MOTOR EARTHING CONDUCTOR
△04
MOTOR △03
△03
MOTOR EARTHING
GI STRIP 40X5 FOR SKID EARTHING △03

SECTION C-C
SCALE 1 : 15



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L&T - CHIYODA LIMITED

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COMMENTED BY	REVIEWED BY	APPROVED BY	DATE

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DRAWING NO.	DRAWING TITLE	REV
504-LZ-930A/B	CIVIL DRAWING OF DOCK LEVELER △01	01

REFERENCE DOCUMENTS / DRAWINGS

DRG. CATEGORY: (USE TICK MARK) ☐ R ☐ I
SCH. DATE OF SUBMISSION
EIL REVIEW/APPROVAL CODE:
VPTL NO.:

DOCUMENT STATUS
STATEMENT OF SUB: 1st 2nd 3rd
☐ FINAL FOR CODE-1 ☑ AS BUILT
ISSUED FOR
☐ AS BUILT ☐ INFORMATION
☐ CONSTRUCTION ☐ REVIEW
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NAME DICIPLINE SIGN. DATE

REV.	DESCRIPTION	DRN	CHECKED	APPROVED	REVIEWED	APPROVED	DATE
01	ISSUED FOR APPROVAL	SMS	VJ	AK			

OWNER: HINDUSTAN PETROLEUM CORPORATION LIMITED
VISAKH REFINERY MODERNIZATION PROJECT
EPCC-03 PACKAGE FOR RESIDUE UP-GRADATION FACILITY (RUF)

PMC: इंजीनियर्स इंडिया लिमिटेड ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

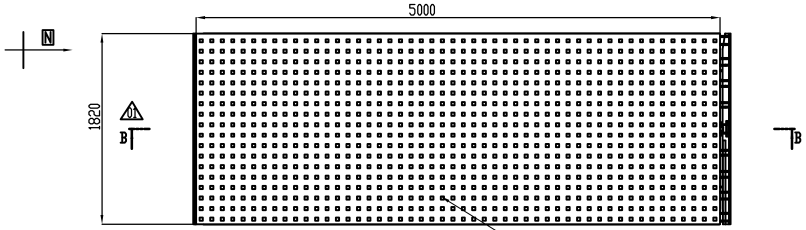
LSTK CONTRACTOR: L&T Hydrocarbon Engineering

DETAILED ENGINEERING CENTER: L&T - CHIYODA LIMITED

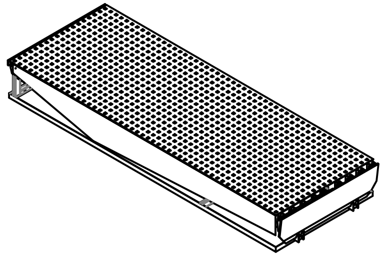
VENDOR :
NIHVA
TECHNOLOGIES
PVT LTD

VENDOR PO NO: REFINARY/75000-82256/VMM
VENDOR DWG. NO: NIHVA_20-21_0157

DATE 31.05.21
SCALE 1:1
DEPT: DESIGN
TITLE: GAD FOR DOCK LEVELER (504-LZ-930A/B) △03
DRN BY: SMS
CHK BY: VJ
APP BY: AK
PROJECT: L&T-RUF
PWO NO.:
DRG. NO.: [illegible]
SH.3 OF5
REV. 04
CAD [illegible] FILE:



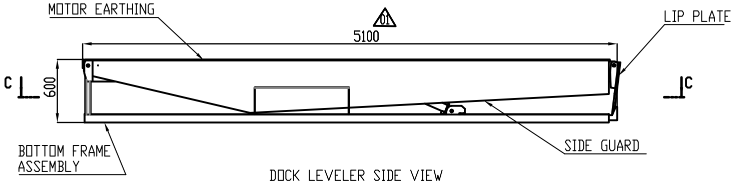
DOCK LEVELER TOP VIEW



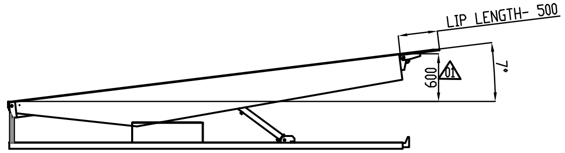
ISOMETRIC VIEW OF DOCK LEVELER

BOM (For Single Dock Leveler) 02

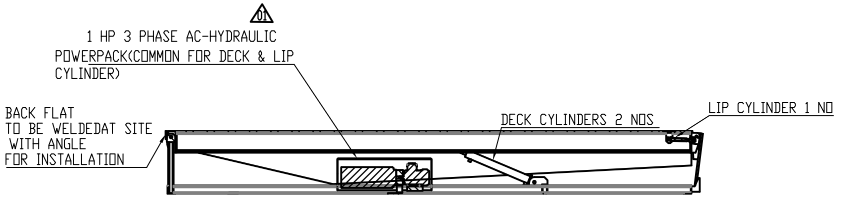
SL NO.	DESCRIPTION	QTY
1	Dock Leveler(1820m x 5000mm)	1No
2	Lip (1820 x500mm)	1No
3	Deck Cylinder	2Nos
4	Lip Cylinder	1No
5	Power Pack(1HP,3Phase)	1No
6	Control Panel	1No



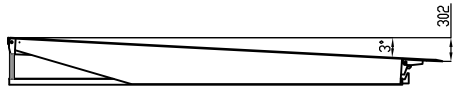
DOCK LEVELER SIDE VIEW



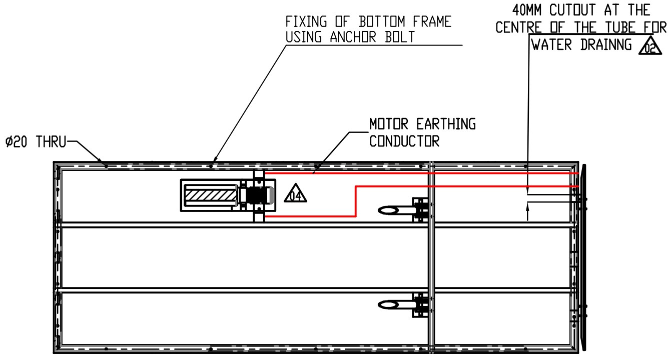
DECK UP HEIGHT - 600 MM


01
1 HP 3 PHASE AC-HYDRAULIC POWERPACK(COMMON FOR DECK & LIP CYLINDER)
BACK FLAT TO BE WELDEDAT SITE WITH ANGLE FOR INSTALLATION
DECK CYLINDERS 2 NOS
LIP CYLINDER 1 NO

SECTION B-B
SCALE 1 : 15



DECK DOWN HEIGHT - 300 MM

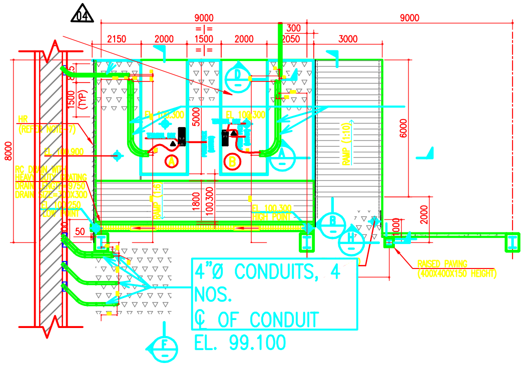

FIXING OF BOTTOM FRAME USING ANCHOR BOLT
40MM CUTOUT AT THE CENTRE OF THE TUBE FOR WATER DRAINNG
02
MOTOR EARTHING CONDUCTOR
Ø20 THRU
04

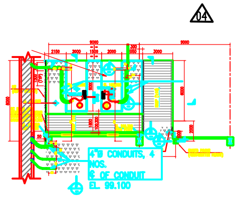
SECTION C-C
SCALE 1 : 15

01

TECHNICAL SPECIFICATIONS OF PIT MOUNTED ELECRO-HYDRAULIC DOCK LEVELER -

1. SIZE OF DOCK LEVELER = 1820mm(W) X 5000mm(L)
2. OVERALL SIZE OF PRODUCT = 1820mm(W)X5100mm(L)X600mm(D)
3. CAPACITY = 6 TON POINT LOAD
4. LIP PLATE SIZE = 500mm(L)X1820mm(W)
5. DECK PLATE = CHEQUERED SHEET 1820mm(W)x5000mm(L)
6. DECK & LIP UP=DOWN OPERATION BY HYDRAULIC CYLINDER
7. DECK UP HEIGHT= (+)600 MM
8. DECK DOWN HEIGHT= (-)300MM 02
9. HEIGHT OF DOCK LEVELER FROM ROAD LEVEL= 900 mm,
10. OPERATION CONTROL BY PUSH BUTTONS ON CONTROL PANEL MOUNTED ON STAND NEAR DOCK LEVELER
11. ELECTRICAL SUPPLY OF EQUIPMENT= 415 VOLT 3 PHASE AC , 50HZ
12. PAINT = FIROZI BLUE
13. MATERIAL OF CONSTRUCTION = MILD STEEL AS PER IS STANDARD
14. MANUFACTURING STANDARD OF DOCK LEVELER =EN1398
15. ELECRICAL MOTOR COMPLIANCE ZONE 1-IIC-T3 HAZARDOUS AREA
16. PIT SIZE = 1870 mm(W)X5100mm(L)x600 mm(D)
17. DOCK LEVELER FIXING IN PIT BY ANCHOR BOLT .
18. REQUIRED PIT & ALL CIVIL WORK FOR MOUNTING OD DOCK LEVELER IS IN LTHE SCOPE .
REFER PIT DRAWING FOR CONSTRUCTION OF PIT B016-RUF-LT-504-MP-DW-NTPL(1)-13005
19. HYDRAULIC CYLINDERS & POWERPACK IS DESIGNED FOR LIFTING BARE DOCK ONLY.
20. MOTOR RATING=1HP 3 PHASE AC-HYDRAULIC
21. All ELECTRICAL EQUIPMENTS SHALL HAVE PAINT SHADE RAL 7031 02
22. ALL FLAME PROOF CONTROL PANEL 375X375X230MM 05
23. WEIGHT= APPROX 2500KG EMPTTY & 8500KG OPERATING 02
24. WE WILL PROVIDE FRP CANOPY FOR LOCAL CONTROL PANEL. 02
25. FOR DOCK LEVELER PACKAGE LTHE SCOPE SHALL BE LIMITED TO PROVIDE SINGLE POINT SUPPLY (3 PHASE, 415 V,50 HZ) TO ISOLETOR PANEL LOCATED AT GRADE LEVEL. FURTHER ALL REQUIRED DISTRIBUTION SHALL BE IN VENDOE SCOPE. 03
26 REFER FOLLOWING TAG NUMBER FOR ELECTRICAL ITEMS.
MOTOR-504-LZ-930A-M1 & 504-LZ-930B-M1.
PANEL-LCP-504-LZ-930A & LCP-504-LZ-930B 03
27. LOCAL PANEL IS SUITABLE FOR ZONE1,IIC,T6 05





504-LZ-930A/B	CIVIL DRAWING OF DOCK LEVELER 01	01
DRAWING NO.	DRAWING TITLE	REV

REFERENCE DOCUMENTS / DRAWINGS

DRG. CATEGORY: (USE TICK MARK) R I
SCH. DATE OF SUBMISSION
EIL REVIEW/APPROVAL CODE:
VPTL NO.:

DOCUMENT STATUS
STATEMENT OF SUB: 1st 2nd 3rd
☐ FINAL FOR CODE-1 ☑ AS BUILT

ISSUED FOR
☐ AS BUILT ☐ INFORMATION
☐ CONSTRUCTION ☐ REVIEW

SIGN: DATE:

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NAME DICIPLINE SIGN. DATE

REV.	DESCRIPTION	DRN	CHECKED	APPROVED	REVIEWED	APPROVED	DATE
01	ISSUED FOR APPROVAL	SMS	VJ	AK			

OWNER HINDUSTAN PETROLEUM CORPORATION LIMITED
VISAKH REFINERY MODERNIZATION PROJECT
EPCC-03 PACKAGE FOR RESIDUE UP-GRADATION FACILITY (RUF)

PMC : इंजीनियर्स इंडिया लिमिटेड ENGINEERS INDIA LIMITED
(A Govt. of India Undertaking)

LSTK CONTRACTOR: L&T Hydrocarbon Engineering

DETAILED ENGINEERING CENTER: L&T - CHIYODA LIMITED

L&T - CHIYODA LIMITED
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VENDOR : NIHVA TECHNOLOGIES PVT LTD
VENDOR PO NO: REFINARY/75000-82256/VHM
VENDOR DWG. NO: NIHVA_20-21_0157
DATE 31.05.21
SCALE 1:1
DEPT: DESIGN
TITLE: GAD FOR DOCK LEVELER (504-LZ-930A/B) 03
DRN BY: SMS
CHK BY: VJ
APP BY: AK
PROJECT: L&T-RUF
PWO NO.:
DRG. NO.: [illegible]
SH.4 OF5
REV. 04
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